In other words, we take the inner product of <φ| with |A> (the result is a complex number) and multiply it by the ket |ψ>. The bra-ket notation is so efficient that it practically forces the definition on us. That was the genius of Paul Dirac. It's easy to prove that the outer product can also act on bras:

$$\langle B| \ \ |\psi\rangle\langle\phi| \equiv \langle B|\psi\rangle \ \ \langle\phi|.$$

A special case is the outer product of a ket with its corresponding bra, |ψ><ψ|. Assuming that |ψ> is normalized, this operator is called a *projection operator*. Here is how it acts:

$$|\psi\rangle\langle\psi| \ \ |A\rangle = |\psi\rangle \ \ \langle\psi|A\rangle$$

Note that the result is always proportional to |ψ>. A projection operator can be said to project a vector onto the direction defined by |ψ>. Here are some properties of projection operators that you can easily prove (remember that |ψ> is normalized to 1):

- Projection operators are Hermitian.
- The vector |ψ> is an eigenvector of its projection operator with eigenvalue 1:

$$|\psi\rangle\langle\psi| \ \ |\psi\rangle \ = \ |\psi\rangle$$

- Any vector orthogonal to |ψ> is an eigenvector with eigenvalue zero. Thus, the eigenvalues of |ψ><ψ| are all either 0 or 1, and there is only one eigenvector with unit eigenvalue. That eigenvector is |ψ> itself.
- The square of a projection operator is the same as the projection operator itself: